2270 Financial Modeling: Formulae

Cash Flows.
Present value of a series of cash flows:

$$PV = \sum_{t=1}^{N} \frac{CF_t}{(1+r)^t}$$

Future value of a series of cash flows:

$$FV_T = \sum_{t=1}^{T} CF_t(1+r)^{T-t}$$

Net present value of a series of cash flows:

$$NPV = CF_0 + \sum_{t=1}^{N} \frac{CF_t}{(1+r)^t}$$

Internal rate of return (IRR):

$$CF_0 + \sum_{t=1}^{N} \frac{CF_t}{(1+IRR)^t} = 0$$

Present Value of Annuity

$$PV = \frac{CF}{r}\left(1 - \frac{1}{(1+r)^n}\right)$$

Bonds.

$$Coupons = \frac{Coupon\ Rate \times Face\ Value}{Number\ of\ Coupon\ Payments\ per\ Year}$$

Bond price

$$P = \sum_{t=1}^{N} \frac{C_t}{(1+r)^t}$$

Bond Duration

$$D = \frac{1}{P}\sum_{t=1}^{N} \frac{tC_t}{(1+r)^t}$$

Portfolios.

Beta

$$\beta = \frac{cov(R_t^{ex}, R_t^{M,ex})}{var(R_t^{M,ex})}$$

Sharpe Ratio

$$SR = \frac{E(R - R^f)}{\sigma(R - R^f)} = \frac{Avg.Excess\ Return}{Volatility}$$

Information Ratio

$$IR = \frac{E(R - R^b)}{\sigma(R - R^b)}$$

High Water Mark

$$HWM_t = \max_{s \leqslant t} P_s$$

Drawdown

$$DD_t = (HWM_t - P_t)/HWM_t$$